

Phase 0.3 — Triangle-Solving Drills

Type: Tool drill, no mission narrative. **Target time:** ~30 min. **Prerequisite:** 0.0 Slide rule basics, 0.1 Trig/log table interpolation.

Much of hand orbital mechanics — tracking geometry, Gauss's method, range- and-bearing problems — reduces at some point to careful triangle solving. This drill covers the three tools you'll lean on repeatedly: law of cosines, law of sines, and the ambiguous SSA case (which trips up more people than it should, and is worth meeting here rather than mid-problem later).

Law of cosines

$$c^2 = a^2 + b^2 - 2ab \cdot \cos(C) \quad (\text{SAS: two sides + included angle} \rightarrow \text{third side})$$
$$\cos(C) = (a^2 + b^2 - c^2) / (2ab) \quad (\text{SSS: three sides} \rightarrow \text{any angle})$$

Law of sines

$$a/\sin(A) = b/\sin(B) = c/\sin(C)$$

The ambiguous case (SSA)

Given two sides and a non-included angle, there can be **zero, one, or two** valid triangles. If the given angle is acute and the side opposite it is shorter than the other given side, check both the arcsine solution and its supplement ($180^\circ - \text{that angle}$) — both may produce a valid triangle. This is a genuine geometric ambiguity, not a mistake to avoid; real tracking data resolved it with a second observation or physical reasoning about which solution made sense.

Drill 1 — SAS

Sides $a = 42.3$, $b = 57.8$, included angle $C = 63.5^\circ$. Find side c .

Drill 2 — SSS

Sides $a = 35.0$, $b = 48.0$, $c = 61.0$. Find angle C (opposite the longest side).

Drill 3 — ASA

Angle $A = 41.0^\circ$, angle $B = 67.0^\circ$, side $a = 28.5$ (opposite A). Find angle C , side b , and side c .

Drill 4 — SSA (ambiguous case)

Side $a = 22.0$, side $b = 30.0$, angle $A = 25.0^\circ$ (opposite side a). Find all valid solutions for angle B , and complete each triangle (angle C and side c).

Worksheet

	Drill	Given	Working	Answer
1		$a=42.3, b=57.8, C=63.5^\circ$		$c =$
2		$a=35.0, b=48.0, c=61.0$		$C =$
3		$A=41.0^\circ, B=67.0^\circ, a=28.5$		$C= __\ b= __\ c= __\$
4a		$a=22.0, b=30.0, A=25.0^\circ$		$B= __\ C= __\ c= __\$
4b		(second solution)		$B= __\ C= __\ c= __\$